

MODULE 1

SL	IMPORTANT QUESTIONS
1	According to structural capability of robot how are robots classified ? Explain with examples.
2	State 3 laws of robotics & applications.
3	Give different types of mobile robot with an application.
4	What are the three degrees of freedom associated with the arm and body motion of a robot?
5	Explain the four basic robot configurations classified according to the coordinate system.
6	Sketch a robot wrist and indicate 3 DOF associated with the wrist.
7	Briefly explain the dynamic characteristics of robots.
8	What is the work envelope of a robot? Sketch two views to indicate the work envelope of a Cartesian robot.
9	Discuss in detail the anatomy of a manipulator.
10	Give advantages and disadvantages of any three configurations with neat sketches in a robotic manipulator.
11	Illustrate the concept of degrees of freedom in the context of robotics with suitable examples. Sketch a robotic wrist and give its degrees of freedom.
12	Describe in detail the applications of robots.
13	List and explain the features of four basic wheel types used for a robotic application.
14	List and define the factors influencing the design of a robot.
15	Explain the anatomy of a robotic manipulator.
16	Describe the various joint configurations possible in a robotic manipulator.
17	Explain the principle of operation of various grippers used in robotic systems.
18	Explain the general features of a legged robot and differentiate the terms static and dynamic stability in the context of a legged robot.
19	Illustrate degrees of freedom in context of robotics with a suitable example.
20	What are end-effectors ? List various types of end-effectors.
21	Sketch and explain any three robotic configurations.
22	List any 2 types of joints with suitable sketches.

23	Explain the following: • Wheeled mobile robots • Legged mobile robots • Aerial robots • Underwater robots
24	What are actuators, sensors, controllers in robotics? Give suitable examples.

MODULE 2

SL	QUESTION
1	Define the term sensor, and differentiate the same from the transducer.
2	Compare CCD camera with CMOS Camera.
3	Discuss any two position sensors.
4	Explain the significance of H- Bridge & PWM in a motor driver.
5	Discuss the different sensor traits that should be taken into account when selecting a sensor for a robotic application.
6	Explain in detail with suitable diagrams, graphs and example the working of ON -OFF controller.
7	Demonstrate the control system by means of a block diagram.
8	Differentiate between servo motor and stepper motor.
9	A mobile robot is designed for unidirectional motion with constant velocity. Design a mechanism to make the robot move in forward and reverse directions with controllable speed and suggest the sensors and motion control drives required to achieve the same.
10	Differentiate Stepper Motor and Servomotor concerning the construction and applications.
11	Describe the characteristics of sensors. Explain any one sensor used in robotics with the help of neat diagrams.
12	Explain the working of the touch sensor used in robotic applications with a neat sketch.
13	Define the terms Dynamic range, Linearity, and Resolution for a Sensor.

14	Position sensors are not preferred to obtain the velocity and acceleration, Give reason.
15	Explain the various sensor characteristics to be considered in the choice of the sensor for robotic application.
16	Explain the working principle of Hall effect sensor and any one application of the same.
17	Compare the performance characteristics of a hydraulic and pneumatic actuator.
18	Explain with a neat circuit diagram the working principle behind a permanent magnet stepper motor
19	Differentiate between servo-motor and stepper motor.
20	Explain any three sensor performance measuring parameters.
21	What is an actuator in the context of robotics? What are the two common types of actuators used in robotics and describe how they work.
22	Explain the significance of the H-Bridge's four switches in controlling motor direction?
23	How are sensors classified ? Explain in detail with suitable examples.
24	What are the characteristics that need to be considered on choosing a sensor for various purposes ?

MODULE 3

SL	QUESTION
1	Give the importance of image processing analysis in robotic vision applications.
2	Define the terms Steerability, and Maneuverability.
3	Differentiate between Car-like mobile robot and differential drive mobile robot.
4	A point P with coordinates (2, 3, 4) in the reference frame is translated by a vector $(dx, dy, dz) = (1, -2, 3)$. Find the coordinates of point P in new coordinate system.
5	In the context of robot vision give an outline on the following: Pre-processing, Segmentation, Description, Recognition
6	With help of suitable examples, explain the terms Degree of Maneuverability, Degree of Steerability, Degree of Mobility. State the relation between the same.

7	Derive the equation for the representation of a pure rotation about x- axis with suitable figure. Write the equations for representations for rotation about y-axis and z-axis.
8	Give two advantages and two applications of omni directional mobile robots ?
9	Derive the equation for the representation of a pure translation with respect to frame movement in space.
10	Write a short note on the following i. Pre-Processing ii. Segmentation and iii. Recognition.
11	What is an omnidirectional robot? Explain two configurations to set up an omnidirectional robot.
12	A Frame B is rotated 90° about the z-axis, then translated 2 and 4 units relative to the n and o-axes respectively, then rotated another 90° about the n-axis, and finally, 90° about the y-axis. Find the new location and orientation of the frame.
13	Differentiate between holonomic and nonholonomic robots.
14	Write the significance of differential drive, in mobile robotics?
15	Explain the seven stages of robotic vision.
16	Differentiate and explain the working of a CCD and CMOS camera in a robotic vision system
17	List the different co-ordinate frames used in robotic systems and explain the need for mapping or transformation of co-ordinate systems for the same.
18	Frame {1} and {2} have coincident origins and differ only in orientation. Frame {2} is initially coincident with frame {1}. Certain rotations are carried out about the axis of the fixed frame {1}. First rotation about X axis by 45° then about Y axis by 30° and finally about X axis by 60° . Obtain the equivalent rotation matrix 1R_2.
19	The Kinematics of car-like mobile robot is different from differential drive mobile. Justify.

20	Give any two characteristics that distinguish a holonomic mobile robot from a non-holonomic one? How does holonomicity affect the robot's maneuverability?
21	Explain any four types of wheel with suitable sketches.
22	Define the terms Degree of steerability, degree of maneuverability, degree of mobility
23	Derive the Kinematic Model for differential drive robot with a neat sketch.
24	A robot is located at position (2, 3, 4) in a 3D coordinate system. It undergoes a sequence of transformations: • Translation by (1, -2, 3) units. • Rotation around the z-axis by 45 degrees. • Translation by (-3, 1, 2) units. i. Write an equation that describes the motions. ii. Find the final location of the point attached to the frame relative to the reference frame.

MODULE 4

SL	QUESTION
1	What do you mean by Robot Localization? And list out the challenges in localization.
2	State the applications and advantages of SLAM robots.
3	State the differences between Visual SLAM and Graph based SLAM
4	How can you represent the position and orientation of a Robot ?
5	Apply the concept of Kalman filter localization for mobile robot and state the steps with a help of schematic.
6	What is the significance of odometric position estimation in localization? List the challenges of robot localization.
7	Assume, a domestic robot that is supposed to work in home environments. In this case the robot should be able to detect changes in the map due to the rearrangement of the furniture and then perform localization accordingly. Explain how localization can be done without the existence of map.
8	Define SLAM mathematically. Give any two open challenges of SLAM.
9	In the context of robotic map representations describe in depth on decomposition strategies and the associated difficulties.
10	Apply the concept of Accurate Odometry and Error Modelling for a Mobile Robot with a neat sketch.
11	Discuss in detail about the Map representation and list the current challenges in map representation

12	How the probabilistic map based localization can be achieved? And suggest the best filtering method for achieving the same.
13	List and explain the Decomposition strategies used in map representation.
14	Identify mobile robot localization problems.
15	Describe the concept of mobile robot localization with suitable Block diagram.
16	Define the term odometry and explain the various error sources in odometric data.
17	Derive error model for odometric position estimation.
18	Give the mathematical definition of SLAM and elaborate the concept.
19	Compare and Contrast graph-based and particle SLAM
20	What are the open- challenges in SLAM ?
21	Differentiate between Graph based SLAM and Particle based SLAM.
22	Explain the significance of odometric position estimation in robotics ? Derive an error model for the same.
23	Discuss any two decomposition strategies in map representation ?
24	Explain localization and mapping in robotics. What is the significance of SLAM?
25	What are the challenges of mobile robot localization? Explain with example.

MODULE 5

QUESTION

1	Differentiate breadth-first search and depth-first search algorithms.
2	What do you mean by control decomposition?
3	Differentiate between A* Algorithm and Dijkstra's Algorithm.
4	Explain the significance of image processing in robot navigation.
5	In the context of robotic map representations describe in depth on decomposition strategies and the associated difficulties.
6	Compare and contrast the advantages and limitations of BFS and DFS in path planning scenarios, considering the factors completeness, optimality, memory usage and speed.
7	How is Voronoi diagram different from that of visibility graph ? Illustrate with help of suitable sketches.
8	One of the central criticisms of Bug-type algorithms is that the robot's behaviour at each instant is generally a function of only its most recent sensor readings. This can lead to undesirable and yet preventable problems in cases where the robot's instantaneous sensor readings do not provide enough information for robust obstacle avoidance. Explain a technique with suitable diagrams that can overcome the limitation.

9	A neural network can be trained to process sensory data from cameras, Lidar sensor, or other sensors. The network learns to identify objects, obstacles, and landmarks and generates appropriate actions based on this information. Through iterative training and feedback, the neural network can improve its navigation capabilities, adapting to various environments and optimizing its path planning, obstacle avoidance, and decision-making processes of a robot. Summarise the above with suitable sketches.
10	Explain Dijkstra's algorithm with a suitable example.
11	Discuss the concept of Modularity for code reuse and sharing.
12	Give the assumption to be made while applying the Bug algorithm for obstacle avoidance.
13	Explain the bug algorithm for obstacle avoidance in mobile robotics.
14	Compare any two traversal algorithms.
15	Explain the potential field method-based path planning in robotic systems
16	Illustrate the working of Dijkstra's algorithm with a suitable example.
17	Explain the need for control decomposition in a robotic system.
18	With an example, elaborate the process of horizontal decomposition of a robotic control system.
19	Local obstacle avoidance focuses on changing the robot's trajectory as informed by its sensors during robot motion. Give any three methods or algorithm for robot's obstacle avoidance.
20	Explain two types of control decomposition.
21	What is configuration space? What is the significance of path-planning and navigation in robotics? In the context of path planning illustrate the following graph search techniques with suitable examples. • Depth First Search • Dijkstra's Algorithm • A* Algorithm
22	What are the properties of temporal decompositions? Illustrate 4 level - temporal decomposition of a simple navigating mobile robot with a suitable block diagram.
23	Demonstrate how a neural network approach has been integrated into robot navigation.